INCH-POUND

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MILITARY SPECIFICATION

SHEARING MACHINES, METAL SQUARING, MANUALLY OPERATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hand and foot operated squaring shears with equipment and accessories as specified herein.

1.2 Classification. The machines shall be of the following types, styles, and sizes as listed in tables I, II, and III. The type, style, and size to be furnished shall be as specified (see 3.5 and 6.2).

Type I - Bench mounting, hand lever operated
Style A - Gapless frame

Type II - Floor mounting, foot treadle operated
Style A - Gapless frame
Style B - Gap frame

2. APPLICABLE DOCUMENTS

2.1 Governmnt documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, addition, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd., Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal(DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARD

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this docunmt to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent spcified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B11.4 - Shears - Safety Requirements for the Construction Care, and Use. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-A366/A366M - Steel, Sheet, Carbon, Cold-rolled, Commercial Quality. (DOD adopted)

(Application for copies should be addressed to American Society for Testing and Materials, ATTN: Customer Service, 1916 Race Street, Philadelphia, PA 19103 -1187.)

(Non _Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete machine shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 design. The machine shall be new (not a prototype) and one of the manufacturer s current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair. The machine shall be manually operated and designed to accomplish shearing by the downward movement of the crosshead mounted upper shear blade past the stationary lower shear blade.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The machine may contain recalimed materials to the maxinum extent possibl e provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Controls. All mechanical operating controls shall be located convenient to the operator’s work station.

3.2.4 Safety and health requirements, All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform with all requirements of OSHA 29 CFR 1910 and ANSI

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B11.4. If a conflict arises between OSHA 29 CFR 1910 and ANSI B11.4, OSHA 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2).

3.2.4.1 Covers, guards, and other safety devices. Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration the procuring activity shall provide part design data to the supplier of the machine (see 6.2). All special guarding shall meet OSHA 29 CFR 1910. The safety devices shall not interfere with the operation of the machine. All guards shall allow access to the guarded part(s).

3.2.5 Mercury restriction. The machine shall not contain mercury or mercury conpounds nor be exposed to free mercury during manufacture.

3.2.6 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.7 Enviromental protection. During the manufacture, operation, service, transportation, or storage of the machine, and its components materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

3.2.8 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.9 Interchangeability. All parts having the same manufacturer's part nunber shall be functionally and dimensionally interchangeable. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maxinum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

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3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein (see 6.2), the condition and finish of all surfaces shall be in accordance with the manufacturer's commerical practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall be constructed from a ferrous material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque without gear damage, failure, or premature deterioration and wear. All gears shall be manufactured to definite established standards to provide for interchangeability of parts.

3.4 Components

3.4.1 Type I, style A machines. The type I, style A machine shall include a frame,crosshead, bed, bolddown bar, operating mechanism, back gauge, bevel gauge, side squaring gauge, and shear blades.

3.4.1.1 Frame, The frame shall support all components of the machine at a level of rigidity required to perform the shearing operations specified in 3.6. The frame shall be the gapless or straight end style, arranged for mounting. The frame shall be constructed of stress-relieved iron or steel. The frame shall be provided with means for securing and adjusting the lower shear blade to maintain alignment with the upper shear blade.

3.4.1.2 Crosshead. The crosshead shall be constructed of either iron or steel, and arranged to secure and guide the upper shear blade through the shearing cycle. The crosshead shall be provided with adjustable stops for controlling the distance of blade travel. The crosshead travel shall be accomplished on machined bearing surfaces with provision to compensate for wear. Guide ways shall be provided to guide the crosshead through the shearing action and to hold the upper shear blade in alignment The guide ways shall be constructed of either cast iron to tool steel and be directly supported by the franc.

3.4.1.3 Bed. The bed shall be constructed of either iron or steel, stress-relieved, with the top surface finished to 125 microinches or less, and flat to within a tolerance of + 0.005 inch per linear foot. The bed shall be provided with means for attaching and accommdating workpiece positioning gauges as specified herein.

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TABLE 1. Type I - Bench mounting shears, style A, gapless frame.

Sizes and capacities			
Size	Capacity	Workpiece width, inches, not less than	Back gauge range, inches, not less than
6		6	0 through 12
12		12	0 through 12
24		24	0 through 12
36		36	0 through 25

TABLE II. Type II - Floor mounting shears, style A, - gapless frame.

Sizes and capacities				
Size	Capacity		Back gauge, range, inches, not less than	Front gauge range, inches, not less than
	Workpiece thickness, inches, not less than	Workpiece width, inches, not less than		
36	0.0598 (#16 ga)	36	0 through 24	0 through 30
48	0.0598 (#16 ga)	48	0 through 24	0 through 30
52	0.0598 (#16 ga)	52	0 through 28	0 through 30
72	0.0598 (#16ga)	72	0 through 24	0 through 30

TABLE III. Type II - floor mounting shears, style B - gap frame.

Sizes and capacities				
Size	Capacity		Back gauge, range, inches, not less than	Front gauge range, inches, not less than
	Workpiece thickness, inches, not less than	Workpiece width, inches, not less than		
52	0.0598 (#16 ga)	52	0 through 25	0 through 30

NOTE : Requirements are not less than those shown.

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3.4.1.4 Holddown bar. Machines, sizes 12, 24, and 36, shall be provided with a holddown bar for securing the workpiece to the bed. The holddown bar shall be actuated by the machine operating lever to clamp the workpiece prior to shearing on the shear blade down-stroke, and to release the workpiece on the shear blade up-stroke.

3.4.1.5 Operating mechanism, The shearing action of the crosshead shall be accomplished by a hand lever operating mechanism of either the cam and cam-shaft type or the gear segment and pinion-shaft type. The operating mechanism shall be designed to assure that sufficient force is developed by a single stroke of the hand lever to accomplish the shearing performance requirements of 3.6. The hand lever shall be not more than 30 inches in nominal length.

3.4.1.6 Back gauge. The machine shall be furnished with a back gauge to provide an adjustable reference edge for positioning the workpiece parallel with the lower shear blade. The reference edge shall be adjustable by a control with a graduated scale in increments of not larger than 1/32 of an inch for measuring any workpiece cut off width through the back gauge ran ge specified in table I.

3.4.1.7 Bevel gauge. The machine shall be furnished with an adjustable bevel gauge mounted on the bed for positioning the workpiece to any desired angle up to 180 degrees from the lower shear blade. The bevel gauge shall be detachable from the table top.

3.4.1.8 Side squaring gauge. The bed shall be provided with a side squaring gauge for positioning the workpiece at right angles to the lower shear blade. A calibrated scale graduated in increments of not larger than 1/32 inch shall be recessed in the gauge for measuring the workpiece.

3.4.1.9 Shear blades. The crosshead and bed shall be furnished with shear blades. The shear blades shall be high carbon alloy steel, hardened to not less than 57 Rockwell C, and machined for endurance and accuracy. The shear blades for size 6 machines shall be the reversible type, and shall be provided with not less than two cutting edges. Shear blades for size 12, 24, and 36 machines shall be the reversible type, and be provided with not less than four cutting edges. When required, shear blades shall be furnished as specified (see 6.2).

3.4.2 Type II, style A apless frame and style B gap frame. Type II machines shallbe provided with components to Include a frame, bed, end housings, crosshead, holddown bar, operating mechanism, back gauge, side squaring gauge, front gauge and shear blades.

3.4.2.1 Frame. The franc shall include a bed and two end housings joined together to form a single unit for supporting all other components of the machine at a level or rigidity to assure machine performance specified in 3.6. The frame members shall be stress relieved and joined together by interlocked and bolted construction to permit assembly and disassembly.

3.4.2.2 Bed. The bed shall be a stress free construction either cast iron or steel plate, with the top surface finished to 125 microinches or less and flat to within a tolerance of $\pm$ 0.005 inch per linear foot. The bed shall

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support the lower shear blade. The bed shall be provided with adjustments for shear blade clearance and wear. The top surface of the bed shall be provided with a hand slot to facilitate handling the workpiece. The bed shall have provisions for attaching and accommodating the workpiece positioning gauges as specified herein.

3.4.2.3 End housings. The end housings shall be stress-free constructions of either cast iron or steel plate. The type II, style A, machine shall be provided with end housings of the gapless or straight style. The type II, style B machine shall be provided with workpiece clearance gaps recessed in the end housings not less than 18 inches from the lower shear blade. The end housings shall be provided with guide ways for supporting and guiding the crosshead and upper shear blade to accomplish the shearing operations specified herein. The end housings shall have provisions for bolting the machine to the mounting surfaces.

3.4.2.4 Crosshead. The crosshead shall be constructed of either iron or steel, designed to guide the upper shear blade in the shearing action. The crosshead for the type II, style B machine shall be provided with adjustable stops for controlling the distance of blade travel. Crosshead travel shall be accomplished on machined bearing surfaces with provisions to compensate for wear. The crosshead shall be provided with a steel truss rod with tension adjustments for maintaining alignment of the upper shear blade.

3.4.2.5 Holddown bar. The workpiece opening of the machine shall be provided with a holddown bar for securing the workpiece to the bed for shearing. The holddown bar shall accommodate any workpiece size up to and including the capacity of the machine. The holddown bar shall be arranged to apply uniform holddown pressure, preventing workpiece movement, The holddown bar shall be designed to provide an unobstructed view of the shearing acti on from the operator's station when the holddown bar is clamped in position.

3.4.2.6 Type II, style A machine. Type II, style A, sizes 36 and 48 machines shall be provided with a holddown bar that is automatically actuated by the foot treadle, clamping the workpiece prior to shearing on-the shear blade downstroke and releasing the workpiece on the shear blade upstroke. Type XI, style A, sizes 52 and 72 machines shall be provided with a holddown bar of the hand lever operated, self-locking, eccentric type.

3.4.2.7 Type II, style B machines. Unless otherwise specified (see 6.2), the type II, style B machines shall be provided with a holddown bar of the hand lever operated, self-locking, eccentric type.

3.4.2.8 Operating mechanism, The shearing motion of the crosshead shall be controlled by a foot treadle type operating mechanism. The foot treadle shall be extended from a pivoting torque rod which shall be supported at both ends by the end housings of the frame. The operating mechanism shall be provided with linkage between the treadle and torque-rod for controlling the stroke of the upper shear blade. The operating mechanism shall be arranged with springs or other devices required to provide automatic return of the foot treadle to the original operating position when released by the operator. The foot treadle shall not extend past the end housings of the frame by more than 24 inches.

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3.4.2.9 Back gauge. A back gauge shall be furnished on the rear of the machine to provide an adjustable reference edge for positioning the workpiece parallel with the lower shear blade for shearing. The reference edge shall be adjustable along two adjustment rods which shall be marked to indicate the adjustment range in graduated increments on one inch, with sub-graduations of not more than 1/16 inch. The adjustment rod shall be provided with pointers for indicating the distance between the reference edge and lower shear blade, and devices for locking the reference edge in position.

3.4.2.10 Bevel gauge. The machine shall be provided with an adjustable bevel gauge mounted on the top surface of the bed, for positioning the workpiece to any desired angle up to 180 degrees from the lower shear blade. The bevel gauge shall be detachable from the bed.

3.4.2.11 Side squaring gauges. The right and left sides of the top surface of the bed shall be provided with side squaring gauges for positioning the workpiece at right angles to the lower shear blade. A scale, calibrated in one inch increments, with sub-graduations of not more than 1/16 inch, shall be recessed or stamped-directly in the gauge for measuring the workpiece. The scale shall be adjustable to compensate for shear blade wear. When required, a side squaring gauge extension arm shall be furnished as specified (see 6.2).

3.4.2.12 Front au e. Two front gauge support arms, with adjustable workstops, shall be atached to the bed for positioning the workpiece parallel to the lower shear blade. The workstops shall be adjustable in T-slots or grooves, machined throughout the full length of the support arms. Means shall be provided for locking the workstops in position.

3.4.2.13 Slitting gauge. When specified (see 6.2), the type II, style B machine shall be provided with a slitting gauge for guiding the workpiece during slitting operations.

3.4.2.14 Shear blades. The bed and crosshead shall be furnished with a shear blade. The shear blades shall be high carbon alloy steel, hardened to at least 57 Rockwell C, and machined for endurance and accuracy. To provide the shear blades with a face clearance, the crosshead shear blade shall be either machined for a face relief slope or arranged in the guides on a shearing plane of one or more degrees off the vertical shearing plane of the bed shear blade. When required, shear blades shall be furnished as specified (see 6.2).

3.4.2.15 Worklight. When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an off-on switch, anc shall be adjustable to reflect maximum light on the work area. If a built in lighting system is provided, the adjustable light described above shall not be required.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the machine shall meet the requirements of table I, II or 111 for the type, style, and size machine specified.

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3.5.1 Physical size limitations. When' specified (see 6.2), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will tit in its future operating location.

3.6 Performance. The shearing machine shall be capable of shearing ASTM A-366/A366M mild steel of workpiece sizes ranging up to and including capacities specified in table I, II, or 111 for the applicable shear type, style and size required. The machine shall be capable of shearing the specified capacities by a single stroke of the hand lever or foot treadle operating mechanism. The resulting sheared edge of the table or bed cut shall be square in conformance with the requirements of 3.6.1 and tables I, II and III. The workpiece shall be free of visible holddown bar damage.

3.6.1 Shearing accuracy. The machine shall be capable of shearing ASTM A366/A366M mild steel of workpiece size up to the full width capacity of the machine, and not less than 0.0359 inch thick, to precision tolerance as follows:

a. Parallelism - The sheared cut shall be parallel to the sheared (trimmed) edge positioned against the back guage within + 0.003 inch per linear foot OF workpiece width.

b. Squareness - The sheared cut shall be square with a sheared (trimmed) edge positioned against the side gauge within + 0.003 nch per linear foot of workpiece width.

3.6.2 Additional performance requirements. Performance features required in addition to, or In lieu of, the requirements herein shal be as specified and fully described (see 6.2).

3.7 Standard equipment. All standard equipment normally provided with the mnufacturer’s commercial machine shall be furnished.

3.8 Optional equipment. When specified, (see 6.2), optional equipment shall be furnished with each machine as follows:

a. Metal mounting stand for type I machines with not less than 31 inches nominal height.

b. Protractor gauge.

c. Other optional equipment as required.

3.9 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.10 Marking on charts and plates. All words on the machine including all charts, plates, and related attachments shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

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3.10.1 Lubrication plate or chart. When specified (see 6.2), a lubrication plate or chart shall be attached to each machine. If a chart is furnished it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets. Unless otherwise specified (see 6.2), the information provided on the chart or plate shall be as listed below.

Points of lubricant application
Type of lubricant
Servicing interval
SAE nunber or military or Federal specification for each lubricant (if applicable)

3.10.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Contract number or order number
National stock number or plant equipment code
Date of manufacture
Capacity, maximum workplace thickness

3.10.3 Workmanship of the machine and its accessories shall meet all requirements specified herein.

3.11 Technical data. When technical data is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for complience. All items shall met all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with

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all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.5 and all tests in 4.7. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.1. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. Each machine shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 410 to 104° Fahrenheit (F) and 20 percent to 95 percent relative humidity.

4.7.2 Operational test. Each machine shall be operated without shearing, during which time the crosshead, holddown bar, operating mechanism, adjusting mechanisms. and accessories shall be verified during the trial period.

4.7.3 Performance tests. Machine performance shall be tested by shearing 4 workplaces of the maximum size and thickness to determine conformance with requirements of 3.6. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.7.3.1 Shear ca acit The shear capacity of the machine shall be verified by shearing workpiece capacities to comply with 3.6.

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4.7.3.2 Shearing accuracy. Shearing accuracy of the machine shall be verified by shearing workplaces of not less than 0.0359 inch thick and not less than the full width capacity of the machine. The machine shall be operated to perform parallel shearing, and square shearing to comply with 3.6.1.

4.7.4 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 4.7.3 shall be as specified and fully described (see 6.2).

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The squaring shears covered by this specification ar intended for use in the shearing of this gauge metal.

6.2 Acquisition requirements Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, style, and size of the machine required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (inch-pound) (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.4).

g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.4.1).

h. Painting, if different (see 3.3.5).

i. Shear blades, if different (see 3.4.1.9).

j. Holddown bar for type II, style B machines, if different (see 3.4.2.7).

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k. Side squaring gauge extension arm, if required (see 3.4.2.11).

1. Slitting gauge, if required (see 3.4.2.13).

m. Shear blades, if different (see 3.4.2.14).

n. Worklight, if required (see 3.4.2.15).

o. Size and capacity, if different (see 3.5).

p. Specify physical size limitation, if required (see 3.5.1).

q. Additional performance features, if required (see 3.6.2).

r. Specify and fully describe optional equipment if required (see 3.8).

s. Specify applicable repair parts, if required (see 3.9).

t. Lubrication plate, if required (see 3.10.1).

u. Lubrication plate information, if different (see 3.10.1).

v. Nameplate, if different (see 3.10.2).

w. Inspection conditions, if different (see 4.3).

x. First article inspection, if different (see 4.4).

y. Quality conformance inspection, if different (4.5).

z. Additional performance tests, if required (see 4.7.4).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
N/A	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language
N/A	DI-TMSS-80528	Supplemental Data for Commerical Off-The-Shelf (COTS) Manuals	Require all data be in the English Language

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The above DIDs were those cleared as of the date of this specification. The current issue of DOD 501 O. 12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. It should be specified in the remarks section of the DO Form 1423 that all technical data furnished should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le System International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification. Classification changes from the previous issue of this specification are as follows:

MIL-S-23244C	MIL-S-23244D
Type II, style A, sizes 22, 30 42, 60, 96 and 120	Deleted
	Type II, style A, size 52 (new)
Type II, style B, sizes 36, 42, 62 and 72	Deleted

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6.8 Changes from previous issues. Marginal notations are not used in this revision to identity changes with respect to the previous issue due to the extensiveness of the changes.

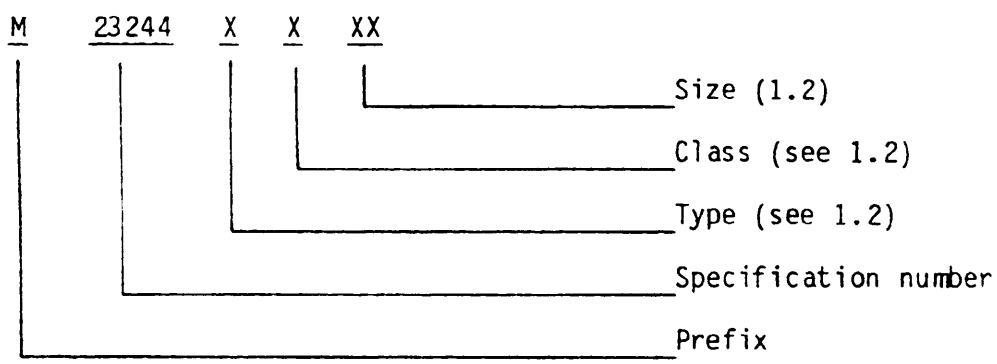
6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the enviroment. in which a machine will operate. It does not limit the 'hazard level of individual machines in an operating enviroment. The procuring activity should analyze the existing hazard level in the proposed operating enviroment and specify additional requirements necessary to integrate this new machine into its future enviroment.

6.10 Trainin Training required by the procuring activity should be provided as specified in the contract.

6.11 Warrant Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed

6.13 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification is created as follows:



6.14 Subject term Key word listing.

Blades
Shearing
Sheet metal
Slitting
Squaring

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Custodian:

Air Force - 99

Review activities:

Army - AL
Navy - SH
Air Force - 84
DLA - GS

User activities:

Navy -MC, SM

Preparing Activity:

DLA - IP

Project Number 3445-0124

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-23244D	2. DOCUMENT DATE (YYMMDD) 91 03 29

3. DOCUMENT TITLE
SHEARING MACHINES, METAL SQUARING, MANUALLY OPERATED

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Industrial Plant Equipment Center ATTN: DIPEC-SSG	b. TELEPHONE (Include Area Code) (1) Commercial: (901) 775-6694 (2) AUTOVON: 683-6694
c. ADDRESS (Include Zip Code) 2163 Airways Blvd. Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 — Previous editions are obsolete — 198-290

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